11. Periodogram-Based Spectral Estimation

Aim

To simulate a noisy sinusoidal signal and estimate its **Power Spectral Density (PSD)** using the **Periodogram Method** in MATLAB.

Objective

- To generate a noisy signal containing two sinusoidal frequencies.
- To estimate the Power Spectral Density (PSD) using the periodogram method.
- To identify the dominant frequency components from the estimated spectrum.

Software Required

- MATLAB R2021a

Code:

```
clc;

clear;

close all;


% Given Parameters

Fs = 2000;         % Sampling Frequency

N = 2048;          % Number of Samples

f1 = 200;          % First Frequency

f2 = 500;          % Second Frequency

SNR = 15;          % Signal-to-Noise Ratio


% Time Vector

t = (0:N-1)/Fs;
```

```
% Original Signal
x = sin(2*pi*f1*t) + sin(2*pi*f2*t);

% Add AWGN Noise
x_noisy = awgn(x,SNR,'measured');

% Periodogram PSD Estimation
[pxx,f] = periodogram(x_noisy,[],N,Fs);

% Plot Noisy Signal
figure;
subplot(2,1,1);
plot(t,x_noisy);
xlabel('Time (s)');
ylabel('Amplitude');
title('Noisy Signal');
grid on;

% Plot PSD
subplot(2,1,2);
plot(f,10*log10(pxx));
xlabel('Frequency (Hz)');
ylabel('Power/Frequency (dB/Hz)');
title('Periodogram Power Spectral Density');
grid on;
```

Output:

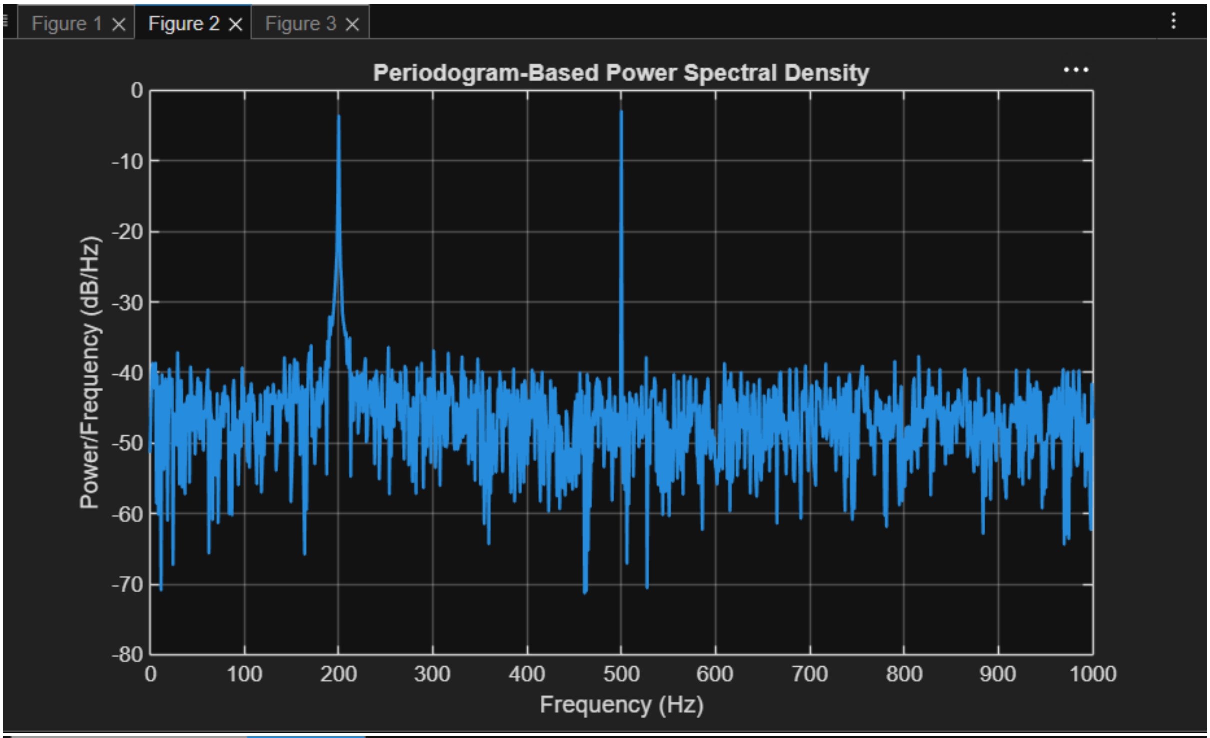
Figure 1
Figure 2
Figure 3
Periodogram-Based Power Spectral Density
Power/Frequency (dB/Hz)
Frequency (Hz)
0
-10
-20
-30
-40
-50
-60
-70
-80
0
100
200
300
400
500
600
700
800
900
1000

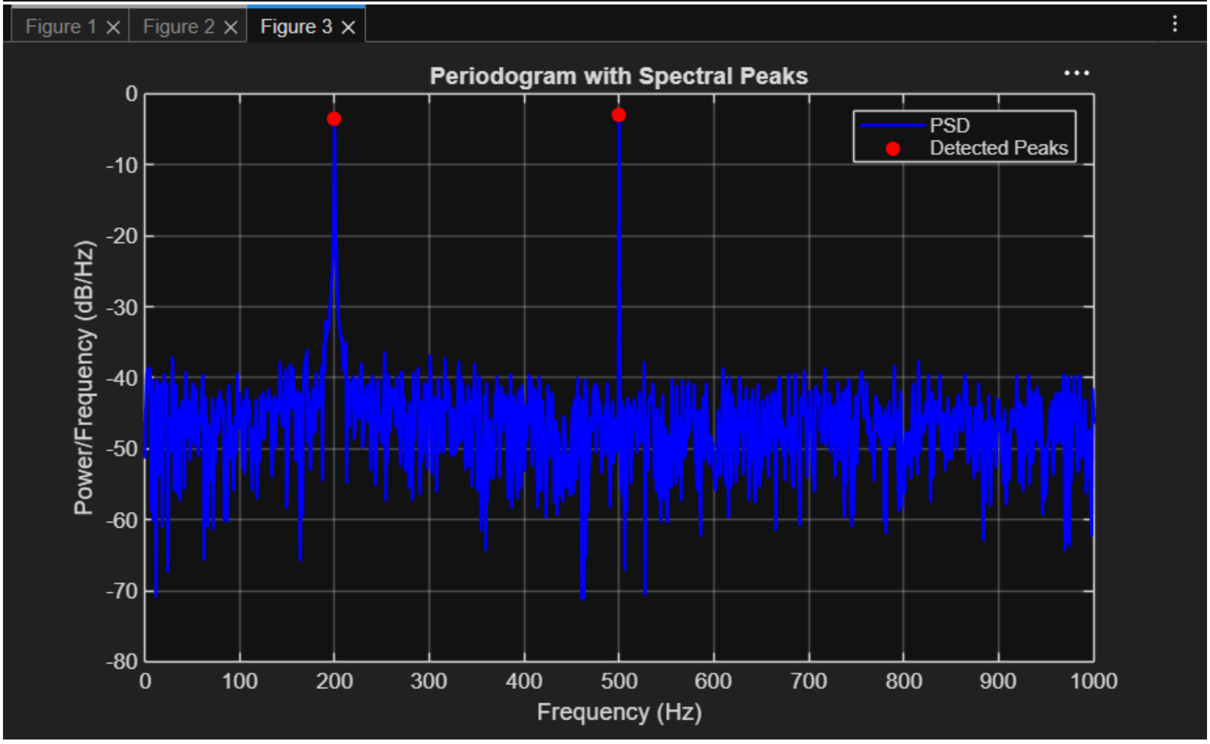
Figure 1
Figure 2
Figure 3
Periodogram with Spectral Peaks
PSD
Detected Peaks
Power/Frequency (dB/Hz)
Frequency (Hz)
0
-10
-20
-30
-40
-50
-60
-70
-80
0
100
200
300
400
500
600
700
800
900
1000

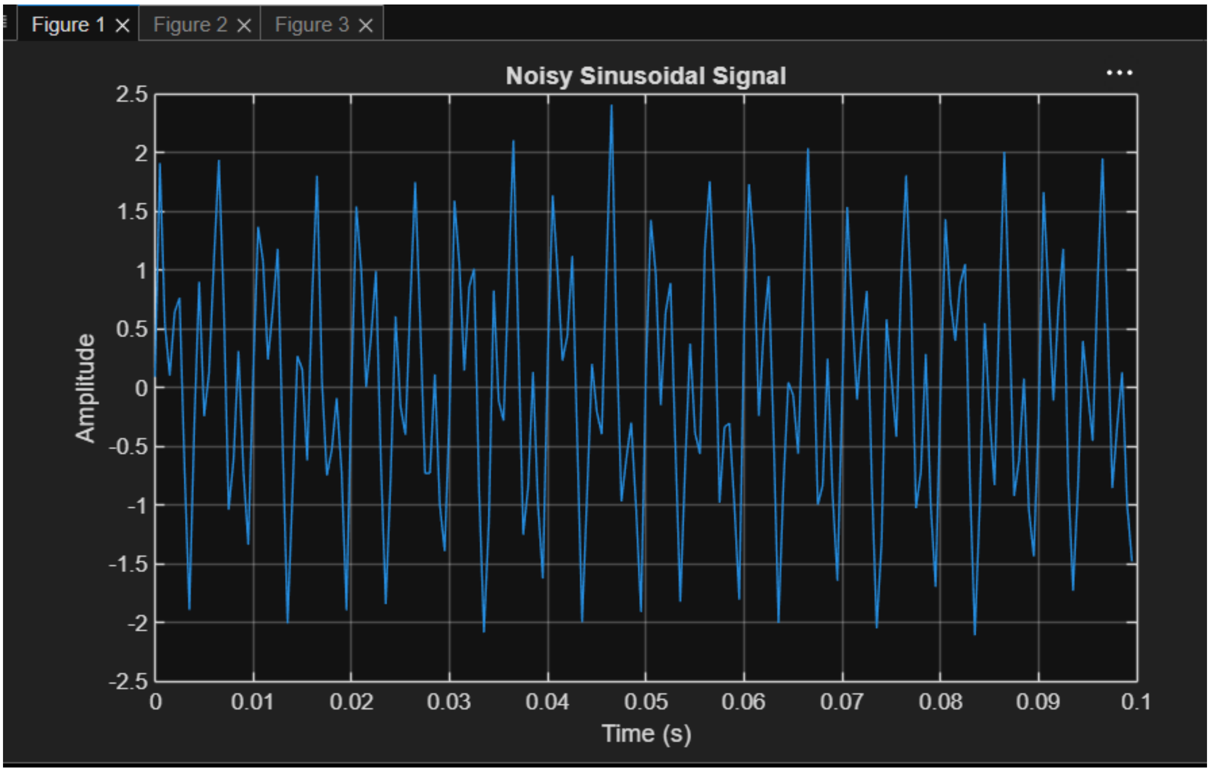
Figure 1
Figure 2
Figure 3
Noisy Sinusoidal Signal
Amplitude
2.5
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
-2.5
0
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08
0.09
0.1
Time (s)